

BIG CHICKIES BRIDGE
Pennsylvania Historic Bridge Recording Project III
Spanning Big Chickies Creek
Lancaster Junction
Lancaster County
Pennsylvania

HAER No. PA-630

HAER
PA-630

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

HISTORIC AMERICAN ENGINEERING RECORD
National Park Service
U.S. Department of the Interior
1849 C St. NW
Washington, DC 20240

HISTORIC AMERICAN ENGINEERING RECORD

BIG CHICKIES BRIDGE

HAER No. PA-630

Location: Spanning Big Chickies Creek, Lancaster Junction, Lancaster County, PA
BMS #: 36723008750202

UTM: Manheim, 18/3811434/4443274

Date of Construction: 1916

Engineer: Firm of Frank H. Shaw

Present Owner: Lancaster County

Present Use: Vehicular traffic

Significance: The tied-through reinforced concrete arch Big Chickies Bridge, built in 1916, is a historically and technologically significant early and rare example in the Commonwealth of Pennsylvania of a rainbow type tied design bridge with diagonal members.

Historian: Richard Vidutis, August 2002

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The fieldwork, measured drawings, historical reports, and photographs were prepared under the direction of Eric DeLony, Chief of HAER. The team consisted of Architects—Todd Croteau, Project Leader (HAER Architect), Roland S. Flores, Field Supervisor (HAER Architect), Marcy Ann Giannunzio (University of Michigan, Ann Arbor), Katherine Marie Kozarek (University of California, Berkeley), Sara Kryda (Illinois Institute of Technology), Jenna Michelle Murphy (University of Detroit-Mercy), Sandra Christine Pires (ICOMOS—Portugal); Dr. Linda S. Phipps and Dr. Richard Vidutis served as project historians under the direction of Dr. Richard O'Connor (Senior HAER Historian). Jose C. Colon (Pennsylvania State University, State College) was the project engineer and Professor Thomas E. Boothby, Ph.D., PE, RA (Pennsylvania State University, State College) was the consulting engineer. Jet Lowe (HAER photographer) took all large format photography. Justine Christianson (HAER Historian) prepared all documentation for transmittal to the Library of Congress.

INTRODUCTION

The Big Chickies Bridge, currently a vehicular bridge, is located in Rapho and Penn Townships, Lancaster County, Pennsylvania and is owned by Lancaster County. The bridge carries Auction Road, a two-lane road, over Big Chickies Creek in an area known as Lancaster Junction. The bridge is situated in a rural area of active farms and scattered residences from the late twentieth century. The setting of the bridge does not have the integrity of a historic district because the farm complexes—found to the northeast and southwest of the bridge—are mixtures of historic dwellings and modern outbuildings. An exact twin of Big Chickies Bridge, the Weaverland Bridge (HAER No. PA-589) was also built in 1916 along Township Route 894 (Quarry Road) over Conestoga River in East Earl Township, Lancaster County in a location called Weaverland.

The Big Chickies Bridge is notable for having been designed by the firm of Frank H. Shaw. Frank H. Shaw, Lancaster County Engineer (1909-1917), along with his brother Percy A. Shaw, Designing Engineer, had a significant impact on reinforced concrete bridge construction in Lancaster County from approximately 1909 to at least the mid-1920s. During that time about forty of their designed concrete bridges were built by Lancaster County. It appears that in a number of cases the firm of Frank H. Shaw also supervised the construction of their designed bridges.

During the course of researching the history of the Weaverland Bridge, the original engineering drawings of the bridge by the firm of Frank H. Shaw were discovered at the Lancaster County Engineer's Office in Lancaster. Special thanks must be given to Mr. Robert Navitski, P.E., Assistant County Engineer, who found the drawings in the office's archives and provided access to the two volumes of Bridge Dockets that contain historic information on reinforced concrete bridges built in Lancaster County from 1908 to 1924. Since both the Weaverland Bridge and the Big Chickies Bridge are dimensionally equivalent, and both were built in 1916, it can be assumed that both were constructed from a set of similar, if not the same, engineering drawings that were standard for both structures.

HISTORY OF THE BIG CHICKIES BRIDGE

Local Neighborhood History

The Big Chickies Bridge sits on the border of Penn and Rapho Township and thus serves both regions as a crossing over Big Chickies Creek.

The Anglicized pronunciation “Chickies” is a rendition of “Chiquesalunga” (meaning “place of crayfish”) from the Delaware Tribe who were part of the Algonquian linguistic group.¹ Early Europeans referred to the Indians living in the vicinity of the creek as belonging to the Chiquesalunga Tribe. Over the years various shortened spellings have been used by the Euro-Americans, who supplanted the Chiquesalunga Tribe, to identify both Big and Little Chickies: “Chiques,” “Chiquis,” “Chikis,” and the now standard “Chickies.”²

The Chickies system of waterways consists of the Little Chickies, Big Chickies and Donegal creeks. They drain the northwestern section of Lancaster County into the Susquehanna River below the Borough of Marietta. Big Chickies rises in Lebanon County and runs in a southwestwardly direction.³

When the early European settlers began arriving in 1716 in the area between the Big and Little Chickies creeks, they found a land barren of old growth forests. It was a condition created by Native American practices of setting yearly fires. Instead of trees the landscape contained stretches of thick-growing hickory and oak saplings and other kinds of underbrush.⁴

In 1846, most of present Penn Township was taken from Warwick Township along with parts of a larger Rapho Township in the vicinity of Mount Hope. Most of the first settlers in the township were of Germanic and Swiss ancestry who began arriving in the area around 1735, settling along the foot of the Furnace Hills. Their landholding extended southward toward what is now White Oak (settled in 1794). The oldest town in the township is Penryn, which was

¹ <http://www.srbc.net/docs/IndianNamesDataChart.PDF>

² Nadine Peterson and Connie L. Torbeck, Historic Resources Survey and Determination of Eligibility Report, S.R. 4003, Section 002, Bridge Replacement Project, Lancaster County, Pennsylvania. Report prepared by Cultural Heritage Research Services, Inc., North Wales, Pennsylvania, 1995, p. 3.

³ Justin Roddy, Physical and Industrial Geography of Lancaster County, Pennsylvania (Lancaster: The New Era Printing Co., 1916), p. 84.

⁴ Franklin Ellis and Samuel Evans, History of Lancaster County (Philadelphia: Everts & Peck, 1883), p. 758; H.M.J. Klein, ed., Lancaster County, Pennsylvania: A History. Vol. 1 (New York and Chicago: Lewis Historical Publishing Co., Inc., 1924), p. 101.

founded in the 1730s. An important industry developed in Limerock in 1880 that took advantage of the abundant limestone found in the southeast corner of the township. Limestone from the quarries was shipped by railroad to other parts of the country.⁵

Mount Hope, an early post town with a stagecoach stop, also had the Mount Hope Chemical Charcoal Works. This important industry engaged in some of the earliest smelting of iron ore in Lancaster County. The village of Elm, originally called Penn and well known for Molly Plasterer's Tavern, was located where five roads converged and was a meeting place for ironworkers from local forges and furnaces.⁶ In time, Penn Township evolved into a growing suburban area for both Manheim and Lititz Boroughs, as well as for the Lancaster urbanized area.

In 1741, citizens of the northeast portion of Donegal Township petitioned the state to erect a new township. It was acted on favorably and the new township of Rapho was established in May of that year. The name Rapho was derived from the town of Raphoe in Ireland, an ecclesiastical center of the county of Donegal. By 1761, Manheim was laid out and by 1838 was incorporated as a borough. Today Rapho Township is the largest municipality by area in Lancaster County, containing approximately 30,300 acres, 20,000 (66 percent) of which is zoned for agriculture.⁷

The township was settled primarily by Scotch-Irish, who lived in the southern portion of the township, and later by Swiss and Germans, who settled in the northern section. In the late nineteenth century, this exceptionally rich agricultural area produced major crops of corn, wheat, oats, and tobacco.⁸

The earliest map researched of the crossing area at Lancaster Junction was by Joshyua Scott and dates from 1824. Although it indicates no crossing, a J. Snavelly is listed as an owner of land at the place where a road and eventually a bridge was placed; that name appears on maps as late as 1899.

⁵ L.H. Everts and D.J. Stewart, Historical Atlas of Lancaster County, Pennsylvania (Philadelphia: Hunter Publisher, 1875); "Penn Township," <http://www.penn.co.lancaster.pa.us/penn/cwp/view.asp?a=3&Q=467386&pennNav=>

⁶ Ibid.

⁷ Manheim Revisited (Manheim: Manheim Historical Society, 2000), pp. 210-211.

⁸ Everts and Stewart.

The H.F. Bridgens from 1864 was printed four years after the first bridge, an iron structure, was built at the project area. The necessity for the bridge was dictated by the presence of a complex of mills owned by Hy. Snavely on the Penn side of the creek. His property was listed as "Distillery & Grist Mercht. & Saw M." Bridgens' 1864 map and Everts and Stewart's 1875 map both show a millrace and what may have been a dam at the point where the millrace took off southward to cross the road. This enterprising area eventually developed into the village of Lancaster Junction by 1899, which, at least on the Penn Township side, had fifty-one structures including a school and a railroad stop called the Lancaster Junction Station. An interesting detail in the Graves and Steinberger's 1899 map is the fact that the creek was still referred to as the "Chiquesalunga Creek."

HISTORY OF THE BIG CHICKIES BRIDGE

Earlier Crossings

In 1860, Elias McMellen and William Corvin were contracted to build an iron bridge for \$1,088.00 at Henry Snavely's Mill on the road from the Colebrook Road to the Lancaster, Petersburg and Manheim Road between Rapho and Penn Townships. The length of the bridge was 63', the roadway measured 16', and the height from low water was 10'. Viewers were sent in 1855 and 1856 to report on the need for the bridge, and in 1861, inspectors filed their approval with the court on the newly built iron bridge.⁹

In subsequent years, H.S. Kaufman made repairs in 1878 for \$135; in 1883, by Peter Helman for \$148.20; and in 1893, by the original builder, E. McMellen for \$590. In 1909, James Bruder refloored, painted, and lightened the iron bridge for \$173.¹⁰

Selection of the Contractors

The iron bridge was replaced in 1916 by a one span concrete bridge with abutments on the same site as the iron bridge between the Borough of Manheim and Penn Township.¹¹ J. Miller

⁹ Bridge Dockets, Commissioners' Office, Lancaster County Court House (small volume), p. 223. Lancaster Court House, Lancaster City, has two undated volumes of Bridge Dockets distinguished only by size (one small and one large volume), but both have bridges arranged by the name of the creek or river they cross. The numbering system for the bridges is flexible in that they are assigned different numbers throughout the history of the bridge.

¹⁰ Ibid.

Eshleman built it for \$2,276.97.¹² By agreement of the County Commissioners of Lancaster and the Borough and Township authorities, the new reinforced concrete bridge was to be kept up and in repair by the Borough of Manheim and Penn Township.¹³

Description of the Big Chickies Bridge

The firm of Frank H. Shaw designed the reinforced concrete Big Chickies Bridge, which J. Miller Eshleman built in 1916.¹⁴ Although engineering drawings for the Big Chickies Bridge were not found during the research phase of this report, they were unearthed for the Weaverland Bridge—an exact twin of the Big Chickies Bridge—and can be used equally to describe the construction features of both.

An interesting anomaly appears in the described length of the Big Chickies Bridge. Bridge Dockets lists the length of the bridge, it is assumed at the time of construction, as 62'-3" long and 15'-11" wide.¹⁵ Yet, in A.G. Lichtenstein's report in 2000, the bridge was measured at 58' long and 20' wide.¹⁶ Apart from errors in measuring, either then or now, what accounts for this discrepancy of 4' in length and 5' in width is not known.

The 58' arch section of the bridge consists of two parallel arches with vertical suspenders from which a concrete slab deck is hung. Within the arch's five panels diagonal suspension ribs (cross beams) are employed, a very unusual feature for a bow arch bridge and one that made the Shaw version distinguishable from the Marsh design. The use of these diagonal elements may have been incorporated into the design in order not to infringe on Marsh's 1912 patent for the Rainbow bridge.

The rebar system of bars and stirrups shows that the four vertical ribs hang from the arch and are tied into four transverse girders onto which the slab roadway is placed.

¹¹ Pennsylvania Historic Bridge Inventory & Elevation, BMS# 36723008750202, Big Chickies Bridge II, prepared by A.G. Lichtenstein & Associates, Inc., Paramus, New Jersey, 2000, erroneously states that the bridge was built in 1922. Data recorded in the Bridge Dockets clearly establishes that the bridge was built in 1916, the same year as its twin at Weaverland over Conestoga River, East Earl Township, Lancaster County.

¹² Bridge Dockets, (small volume).

¹³ Ibid.

¹⁴ Robert Navitski, P.E., Assistant County Engineer, found the drawings in the archives of the Lancaster County Engineer's Office, Lancaster Court House, Lancaster City.

¹⁵ Bridge Dockets, (large volume), p. 48.

¹⁶ Pennsylvania Historic Bridge Inventory & Elevation.

The abutments, piers, wing walls, coping, and floor of the bridge are of concrete.¹⁷ Both ends rest on concrete abutments. The abutments are U-shaped and tied together by reinforced concrete ties. Heavy concrete capped end posts decorate both ends of the main section. Galvanized piping, perhaps functioning as a handrail and guard, extend through the arches.

Much like Percy Frank's Conestoga Bridge at Rupp's Mill discussed above, the Big Chickies Bridge apparently provided the maximum opening needed over Big Chickies Creek at the Lancaster Junction for water to flow under the rainbow arch. Bridges over small rivers or creeks with small apertures can become sizeable obstructions to a river's flow, often forming a weir of driftwood blocking the flow through the apertures. The Big Chickies Bridge was considered long enough to allow the width of the creek to flow freely. Perhaps the history of flooding along Big Chickies Creek in that region of Lancaster County did not necessitate an extension to the bridge as was the case of the twin of Big Chickies Bridge at Weaverland. There the rainbow arch was not modified to produce a single span to cross the approximately 75' wide river but there was an extension.

There are two possible reasons for the situation at the Weaverland crossing, either: 1) a 75' girder slab was considered impractical as an engineering solution or an economic one; or 2) the rainbow arch bridge with diagonal elements was one of a number of readily available and previously designed structures that Shaw offered as solutions for particular crossings and budgets. In comparing the costs of both bridges—the Weaverland Bridge at \$3,630.19, and the Big Chickies Bridge at \$2,276.97—the difference in price for the Weaverland structure was an extra \$1,353.22 for a slab girder section about 18'-8" long resting on a pier and a concrete abutment that extended the bridge roadway southward and allowed for water overflow.

DEVELOPMENT OF THE REINFORCED CONCRETE BRIDGE

In order to understand the Big Chickies Bridge's place in the history of reinforced concrete bridge development, it is necessary to discuss some major phases in the history of reinforced concrete bridge design and construction.

¹⁷ Bridge Dockets, (large volume), p. 154.

The reinforced concrete bridge was first developed in Europe, but it was slow to be accepted in America. In 1876, Jean Monier of France invented the first concrete metal arch by imbedding wire netting in the concrete of the arches. In 1884, R. Wunsch of Hungary, invented the Wunsch system of concrete metal arch building. It consisted of an arched lower and straight upper member of metal imbedded in concrete and connected with vertical members deeply imbedded in piers and abutments. In 1892, Joseph Melan, of Austria-Hungary, developed the concrete metal arch known as the Melan system, which was patented in the United States in 1893.¹⁸ The Melan system consisted of arched ribs filled with concrete and rigidly connected to abutments, beams and girders. Melan bridges were built throughout Europe and the United States (where as many as twenty-seven had been constructed by 1899).¹⁹

In 1894, a German born engineer working in America by the name of Fritz von Emperger, claimed that concrete and steel could be used successfully in the construction of bridges.²⁰ Tests in Austria proved that such a combination would indeed produce a strong structure. The faith in the miraculous qualities of this combination of materials lead to the belief that steel would be permanently preserved if set in concrete and that the bridge would actually strengthen by degree.²¹ In 1902, Daniel B. Luten, a prominent American designer of reinforced concrete bridges, described the advantages of bridges built from such materials:

A concrete arch, properly constructed, is practically indestructible. As it grows older it grows stronger: it is not affected by weather; it requires neither painting nor repairing; it has no plank floor system to be continually out of order; it is made of materials that can usually be purchased in the immediate vicinity, and labor as well as materials may be, to a large extent, the home products of the town or county; and it is a handsome bridge structure—for not only is the arch form especially adapted to artistic treatment, but concrete is a material that can be molded into many forms of ornament at small cost.²²

¹⁸ Carl Condit dates the patent to 1894, see American Building: Materials and Techniques from the First Colonial Settlements to the Present (Chicago: The University of Chicago Press, 1968), p. 174.

¹⁹ Edwin Thacher, "Concrete-Steel Bridge Construction," Engineering News XLII, no. 12 (September 21, 1899): 181.

²⁰ Fritz von Emperger, "The Development and Recent Improvement of Concrete-Iron Highway Bridges" Transactions, American Society of Civil Engineers Vol. XXXI, 1894.

²¹ Thacher, p. 179; Daniel B. Luten explained that since steel gives off carbonic acid, which is the cause of rust, "concrete by its absorption of carbonic acid gas grows continually stronger with age, and the suitability of concrete-steel for bridge construction becomes most evident; as traffic grows heavier, the bridge grows stronger, quite a contrast from any other form of bridge erected." Daniel B. Luten, "Concrete-Steel Bridges," Municipal Engineering XXIII, no. 3 (September 1902): 156.

²² Daniel B. Luten, "Highway Bridges of Concrete," Municipal Engineering XIX, no. 6 (December 1900): 388.

In 1904, M.A. Considere designed and built the first recorded example of a reinforced concrete bowstring bridge in France that appears to have been directly based on a metal bowstring bridge. In America, the first known example to be built was designed by Howard M. Jones, Engineer of the Cumberland River Bridge Commission in Nashville, Tennessee, who considered European concrete bridges to be too complicated. He produced a bridge with concrete trusses without diagonals as deck spans on the approaches to the bridge, and placed the deck above the arches that rested on existing piers.²³ By 1912, Marsh received a patent for his reinforced concrete fixed through arch "Rainbow" bridge. Unfortunately, Marsh's rainbow design received limited acceptance and never became an industry standard. Nevertheless, variations of the Marsh design, in particular the tied through arch, can be found throughout the country, two of which were built in Lancaster County.

The beginning of the twentieth century marked the advent of construction with reinforced concrete in the United States. Engineers were fascinated with its potential as a building material and believed it to be practically indestructible, making bridges nearly maintenance and replacement free.²⁴ Indeed, this relatively new construction material provided exceptional compression strength at a low cost compared to metal, which always needed the upkeep of painting and even replacement of its metal parts. During the first two decades of the twentieth century, concrete met the needs of municipalities that were looking for replacement bridges that offered economy, strength, and attractiveness.²⁵ In 1908, the best example to spur those views in Pennsylvania was the construction of Philadelphia's Walnut Street Bridge. The 233' long concrete structure set an example in its size, beauty and park setting for all engineers of the day who were fascinated with concrete as a construction material.²⁶

For engineers building with concrete during this time period, the main problem was how to use the qualities of concrete's compressive strength along with its low-tension resistance irrelevant to a structure. Influential engineering writers F.E. Turnaure and E.R. Maurer advised

²³ Frances A. Jones, "Benson Street Concrete Bowstring Bridge," HAER No. OH-50, Historic American Engineering Record (HAER), National Park Service, U.S. Department of the Interior.

²⁴ Thacher, p. 180.

²⁵ John Lyle Harrington, "Recent Developments in Bridge Superstructures," Proceedings of the Engineers' Society of Western Pennsylvania 46, no. 3 (Mar. 1930): 63.

²⁶ Helen P. Ross, "Walnut Lane Bridge," HAER No. PA-504, Historic American Engineering Record (HAER), National Park Service, U.S. Department of the Interior, 1998; revised and expanded by Justin M. Spivey, 2001.

engineers that the solution lay not in choosing between concrete and steel, but in using both together.²⁷ Although steel was expensive, it had high tensile strength especially in the form of bars, but needed to be made more heat resistant and durable. Concrete seemed to support steel in ways that made its expense affordable; for example, it greatly slowed steel's corrosive process while itself being durable and fireproof. Concrete was much cheaper because, unlike steel, which had to be produced at a mill, it could be mixed at the work site. Turnaure and Maurer compared current prices for concrete and steel and concluded that encasing steel rods, beams, or columns in concrete would significantly alter the actual cost per unit of strength in bridge building. The savings would come from the use of steel reinforcing rods instead of using fabricated steel members. By the 1930s, it was becoming clear that concrete structures suffered from their own litany of problems, including discoloration, streaking, and cracks. Poor workmanship made it difficult to correct errors in installation.

The Rainbow Arch Bridge

The type of arch bridge employed over Big Chickies Creek at Lancaster Junction was designed by James B. Marsh (1856-1936) who received a patent for it in 1912. The patent application stated that the object of the invention was "to construct an arch bridge of reinforced concrete in such a manner as to permit a limited amount of expansion and contraction both of the arches and of the floor." James B. Marsh's patented design used rainbow arches that would expand and contract along with the bridge floor under varying conditions of moisture and temperature. Marsh built hundreds of these bridges in the 1910s to 1930s, mostly throughout the Midwest. Another feature that was attractive was the completely flat floor, in contrast to older bridges that may have had a hump in the middle. There are only a few remaining bridges of this type in the United States.

James B. Marsh, a civil engineer, from Des Moines, Iowa, was an American pioneer in designing bowstring concrete bridges. In the mid-1890s, Marsh began experimenting with the use of concrete in bridge construction. He produced two basic bridge designs where the arch

²⁷ F.E. Turnaure and E.R. Maurer, Principles of Reinforced Concrete Construction 2nd rev. ed. (New York: John Wiley & Sons, 1913), pp. 4-5.

risers above the deck, one for a fixed and one for a tied arch. On August 6, 1912, he received a patent (United States Patent number 1,035,026) for his innovative design for the fixed arch.²⁸ The fixed arch design contained wear plates at the points where the bridge floor came into contact with the beams and abutments. In contrast to the tied arch, which rests on the abutments, the fixed arch continues below deck level and is attached to the abutments. The tied arch allows construction even if the ground is not solid enough to deal with the horizontal forces produced by the arches as they spread their loads towards the abutments. Rather than relying on the foundation to restrain the horizontal forces, the girder itself "ties" both ends of the arch together. A tied arch bridge rests on abutments and causes a vertical load on the abutments. Both types of Marsh arches allowed for the expansion and contraction of the bridge parts during changes in temperature.²⁹

The tied arch bridge has a number of defining structural aspects: it is used where the supporting rock foundation cannot resist the arch thrust; the horizontal thrust of the arch is taken entirely by the tie; the tie arch is always used as a through span; the deck floor is always carried by hangers; and where there are a multiple series of tied arch spans, they are always a succession of individual tied arches.³⁰

The deck of a tied through arch bridge may be either a slab span or a girder span. The reinforced concrete slab span is one of the simplest and least expensive technologies used in bridge construction. A simple slab bridge can be composed of single or multiple slabs, each of which typically ranges from 20' to 50' in length. In a single slab bridge the slab rests directly on the abutments. If multiple slabs are used, joints are visible where the slabs come together. In a continuous slab bridge, the slab is poured as one unit so there are no visible joints. For longer spans intermediate supports of timber, concrete, or steel in a wide range of shapes are used. Slabs are used for short span bridges where headroom is limited and were a common

²⁸ James B. Marsh, Patent: Reinforced Arch Bridge, Patent No. 1,035,026, Patented August 6, 1912.

²⁹ Historic American Engineering Record (HAER), National Park Service, U.S. Department of the Interior, "Spring Street Bridge," HAER No. WI-37. Prints and Photographs Division, Library of Congress.

³⁰ Ibid.

highway bridge type in the 1930-40s. A simple slab bridge is easily distinguished by its flat underside.³¹

Though similar in appearance to a slab bridge, the concrete girder bridge is distinguished by a series of longitudinal concrete beams on its underside. The girders can be an integral part of the slab, known as monolithic construction, or they can be poured separately. Each girder is typically of a uniform dimension throughout its length. Crossbeams are sometimes placed between the girders to provide extra support. The same supports used for slab bridges are also found on girder bridges. Since less concrete is required in the construction of a girder bridge than in the flat slab, it is typically used for longer spans, such as the one used for the Big Chickies Bridge at Lancaster Junction.³²

Marsh's designs brought a number of innovative features to the building of reinforced concrete through arch bridges that resulted in the elimination of many time consuming tasks involved in traditional concrete construction. In the construction of a Marsh arch design, complete trusses were employed for the reinforcement in the arches. The metal truss work was assembled on the ground and then lifted into place. Angle irons, eight to a side, were attached to the arch and served as the hangers for the deck. Steel trusses were also used for the bottom chords, which supported their own formwork thus eliminating costly and time-consuming formwork arrangements beneath the bridge. Next, large steel I-beams were welded to hangers across the width of the bridge to which formwork attached. At this point, the concrete deck was poured. After the deck concrete had cured, the arches and hangers were cast in concrete.³³

The chief advantages of the Marsh building system for engineer designers, fabricators, and municipal treasurers were ease of construction and economy of resources. Technologically the advantages rested in the ability to construct a reinforced concrete bridge with minimal weight and floor thickness, yet allowing for unlimited headroom below the deck. Since all tension is exerted on the vertical hangers, horizontal web members are eliminated. Finally, the whole span

³¹ Ibid.

³² Ibid.

³³ Ibid.

is protected from the weather by an outer layer of concrete. The end result is an aesthetically pleasing structure that provided a strong and economical crossing.³⁴

Construction of Concrete Bridges in Lancaster County, 1908-1924

In 1908, Lancaster County began replacing its wooden and iron bridges with reinforced concrete spans in response to increasing automobile traffic over less traveled local roads. It is within this context, and the rise of popularity of building with reinforced concrete, that the Big Chickies Bridge is found.

Construction of the Big Chickies Bridge can be better understood in the context of the history of concrete bridge construction in Lancaster County as recorded in the Bridge Dockets (both the small and the large volume) of the Commissioners' Office that are now archived at the County Engineer's Office in the Lancaster County Court House. The two books are retained at the County Engineer's Office as documents of Lancaster County's historic bridge building activities. Unfortunately, the two volumes are suffering some deterioration, and not all the pages appear to be present or intact. For example, it is quite clear that a number of photographs of the bridge structures are missing that were taken the year the bridges were constructed. Nevertheless, the information contained there can be gathered into a revealing table of statistics that show concrete bridge construction by Lancaster County during the years of 1908 to 1924, the height of concrete bridge building in Pennsylvania, when Lancaster County alone built at least twenty concrete bridges on its territory, in addition to whatever number the state had also constructed in the county. The two volumes of Bridge Dockets do not contain concrete bridge information beyond 1924.

Table 1 of statistics included below is for twenty reinforced concrete bridges built from 1908 to 1924. The data was taken from both small and large volumes of Bridge Dockets located at the Lancaster County Court House in Lancaster. The information in the Bridge Dockets is incomplete as it is known that the firm of Frank H. Shaw alone designed at least forty bridges.

³⁴ Ibid.

Table 1 shows that the builder of the Big Chickies Bridge concrete bridge, J. Miller Eshleman, built his first concrete bridge in 1916. In 1920, Eshleman built a two-span girder reinforced concrete over Big Beaver Creek in Strasburg and Providence Townships. He is not mentioned again after building his second bridge.³⁵

The most prolific builders of Lancaster County reinforced concrete bridges were John Brubaker and Paul D. Kauffmann. In all, Brubaker worked on five bridges from 1913 to 1918: In 1913, Brubaker built a concrete bridge over Peters Creek along a road from Wick's Mill to Peach Bottom; in 1914, he repaired a concrete bridge that was built by J.S. McIlvain; in 1916, the Weaverland Bridge; in 1917, a bridge over Big Beaver Creek near Carnargo; and in 1918, together with F.A. Heart, a structure over Pequea Creek along Beaver Valley Pike. The last entry in the Bridge Dockets for Brubaker was for a small job, for which he received \$210, to repair the Weaverland reinforced concrete bridge that he built in 1916.³⁶ Paul D. Kauffmann built three Lancaster County reinforced concrete bridges all in 1917. He built three-spans just over 130' long and about 20' wide were built in West Earl Township over the Conestoga River, and a 131' long single span over Cocalico Creek, which divides Warwick & Ephrata Townships.

Judging from the extant documents found in the Bridge Dockets, the majority of reinforced concrete bridges were built by contractors during the sixteen-year period recorded in the volumes were over Pequea (seven) and Conestoga (five) Creeks.

Table 1.
Reinforced Concrete Bridges Contracted by Lancaster County, 1908-1924³⁷

Concrete Bridge Type	Span/Width (ft.)	Date	Builder	Cost	Township(s) [// = between]	Waterway Crossed	Road carried	Location	Bridge Docket Page	Photo (sm); (lg)
1-span(?)	70/16	1908	Ferro-Concrete Co.	\$3,603.50	// Warwick & Elizabethtown	Hammer Creek	?	B.H. Snaveley's Mill	281 (sm); 308 (lg)	Yes; yes
1-span, girder	63/18	1910	Nelson Merydith Co. (Chambersburg)	\$4,334.57	West Earl	Conestoga River	Farmersville to New Berlin	Burkholder's Mill*	125 (sm)	Yes
2-span	77/20	1910	A. Buchanan (Chambersburg)	\$6,308.99	// Strasburg & W Fallow (Chester Co.)	?	?	Ferguson's Mill	373 (lg)	No

³⁵ Bridge Dockets, Commissioner's Office, (small volume), p. 1.

³⁶ *Ibid*, p. 113.

³⁷ Information in the Bridge Dockets in the small (sm) volume and in the large (lg) is incomplete because some fields were not entered and some photographs have been lost. Where that information was absent, the cells in the table are marked by a "?".

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?-span, girder/slab (?)	?	1911	Heppler & Gilbert (Coatesville)	\$1,107.33	// Coleman & Little Britain	W Branch Octoraro Creek	Pine Grove to Union	?	367 (lg)	Yes
1-span	40/16	1913	John T. Brubaker	\$1,795.15	Fulton	Peters Creek	Wick's Mill to Peach Bottom	Dorsey's Mill	489 (sm)	No
1-span	140/16	1913	Nelson Merydith Co. (Chambersburg)	\$15,738.68	// Martic & Conestoga	Pequea Creek	?	at mouth of Pequea Creek	464 (sm)	No
2-span	100/14	1913	J.S. McIlvain	\$2,502.50	Salisbury	Pequea Creek	?	Amos Hess's Mill	421 (lg)	No
3-span, girder on 2 piers	131/17	1914	J.S. McIlvain; repaired by John T. Brubaker	\$4,365.75	East Earl	Conestoga River	Blue Ball to Reading	Samuel Gehman's Mill; or Aaron Martin's Mill (sm Dock. Bk)	111 (sm); 150 (lg)	Yes; yes
2-span arch	79/19,8	1916	John T. Brubaker	\$3,630.19	East Earl	Conestoga River	Quarry Rd.	Eli Martin's Mill	113 (sm)	?
1-span arch**	62,3/15,11	1916	J. Miller Eshleman;	\$2,276.97	// Rapho & Penn	Big Chickies Creek	Colebrook Rd. to Lancaster-Petersburg Rd.	Lancaster Junction	48 (lg); 228 (sm)	?; ?
girder	28,5 /?	1917	John T. Brubaker	\$1,179.50	Eden	Big Beaver Creek	?	near Carnargo	27 (sm)	No
1-span(?)	131/?	1917	Paul D. Kauffmann	\$5,691.00	// Warwick & Ephrata	Cocalico Creek	?	?	51 (sm); 122 (lg)	No; yes
3-span, girder	131/18,5	1917	Paul D. Kauffmann	\$10,213.00	West Earl	Conestoga River	Talmage Rd.	Kafroth's Mill	170 (lg)	Yes
3-span	134/21,5	1917	Paul D. Kauffmann	?	West Earl	Conestoga River	Farmersville Rd.	Wenger's Mill	?	?
2-span, girder (50' each)	100/?	1918	John T. Brubaker & F.A. Heart	\$10,475.94	// Strasburg & W Lampeter	Pequea Creek	Beaver Valley Pike	?	445 (sm)	No
2-span, girder (?)	?	1920	J. Miller Eshleman	\$10,072.36	// Strasburg & Providence	Big Beaver Creek	New Providence to Lancaster	Mylin's Mill	1 (sm)	Yes
1-span(?) girder steel	?	1920	Pearthree	\$12,447.29	// Martic & Conestoga	Pequea Creek	?	Martic Forge	459 (sm)	No
2-span	?	1921	Solomon Butts & Geo. H. Calder	\$12,990.00	// Pequea & Providence	Pequea Creek	?	near Pequea Valley Tavern	447 (sm)	No
2-span	?	1922-1923	Swangler-Feckler Construction Co. (Lebanon)	\$9,248.77; F.H. Shaw supervised construction (\$1,533)	// Leacock & Paradise	Pequea Creek	?	?	425 (lg)	No
3-span, girder	?	1924	Ferro-Concrete Co.	\$13,104.00	// Leacock & Paradise	Pequea Creek	Paradise to Intercourse	?	426 (sm)	Yes

* Percy A. Shaw's article on this bridge in Engineering Record refers to the location as Rupps' Mill, not Burkholder's Mill.

** Data on the Big Chickies Bridge (an exact twin of the Weaverland Bridge designed by the Shaw firm) were missing from both volumes of Bridge Dockets but are included in this list of reinforced concrete bridges.

Percy A. and Frank H. Shaw

In 1916, Frank H. Shaw (1872-1950), Civil, Hydraulic and Sanitary, and Water Works Engineer, and a member of the American Society of Civil Engineers, had an engineering business located in the Breneman Building at 53 Duke Street, Lancaster City. His home

residence at that time was listed as 206 Ruby Street. Advertisements by Shaw in the Lancaster City Directory for the year of construction of the Big Chickies Bridge indicate a wide range of engineering services, including surveys of subdivisions, cities, boroughs, farm and topographic; and water works, sewage, bridges, railways, tanks, bins, and buildings.³⁸

The Big Chickies Bridge that Eshleman was contracted to build may have been designed by Percy A. Shaw, the brother of Frank H. Shaw. An article, authored by Percy in 1911, on the construction of a reinforced concrete through transverse girder bridge states that he was its engineering designer located in Reading.³⁹ We do not know how long Percy worked with Frank on the firm's construction projects, but the possibility exists that Percy may have been a designer of other bridges created by the company. Because of the name of the firm and the time consuming efforts traveling and selling the firm's wares and reputation, while managing offices in different cities, Frank H. Shaw probably became the better the better recognized of the two brothers.

Frank Shaw moved to Lancaster County around 1905 from Massachusetts and became an inspector of its first concrete bridge built by the Ferro-Concrete Company of Harrisburg in 1908. In 1909, he began advertising his own engineering firm in Lancaster and at the same time was appointed Lancaster County Engineer, a position he held until 1917; from 1911 to 1913, he was also the county's superintendent of water works. After relinquishing his position of County Engineer in 1917, he continued working for the county as a consulting engineer. Frank Shaw's interests were not limited to bridge building but extended to all aspects of reinforced concrete construction. His imprint on the county includes sewer works along Lancaster's Water and Clay Streets, and a water tower now found on the Franklin and Marshall College campus; his water system works are also to be found in Elmira, New York, and Reading, Pennsylvania; he also laid out street railway lines in cities in the East. During his career, he also contracted building works in East Orange, New Jersey (1919-20), Houston (1923-24), and Washington, DC (1935) where

³⁸ Lancaster City Directory (Lancaster: R.L. Polk & Co., Publishers, 1916), p. 112.

³⁹ Percy A. Shaw, "A Reinforced Concrete Through Girder Bridge," Engineering Record 64, no.26 (December 23, 1911): pp. 744-5.

he also had offices. It is for the design and construction of concrete bridges that Frank Shaw, with apparent help from his brother Percy, is best remembered in Lancaster County.⁴⁰

During his tenure as county engineer, Frank (and possibly Percy) designed and developed at least forty reinforced concrete bridges;⁴¹ two of the trusses⁴² are still in service today along with half-a-dozen cantilevered through girder bridges.⁴³ Although Percy's influence and impact on Lancaster County's infrastructure as a designer is subsumed in Frank's reputation, it may have been substantial. Percy noted in his 1911 article that he was the "Designing Engineer" (residing in Reading) of a reinforced concrete through girder bridge at Rupps Mill in Lancaster County.⁴⁴ Therefore, it seems very likely that he may have designed other bridges as well for the Shaw firm. Since Frank was concurrently acting as Lancaster County's County Engineer, Superintendent of Water Works, consulting engineer, supervisor of construction, in addition to running a private engineering construction firm. Also, it can be assumed that Frank Shaw was trying at the same time to sell his company's wares to other regions of the country as seen in the fact that soon after he quit his position of county engineer in 1917, he continued his career as a civil engineer by opening offices in other cities, perhaps in part based on his established reputation in Lancaster County. At least in 1910, Percy worked out of the Reading office and may have designed many of the bridges for which Frank H. Shaw, or the company of Frank H. Shaw, became famous. Frank Shaw's obituary proclaimed a unique place for him in history for his management and supervision of bridge construction projects in Lancaster County. It stated: "the first concrete bridges ever built in Lancaster bear his name as the engineer."⁴⁵

Percy A. Shaw's Ethnographic Engineering Statement

⁴⁰ R.L. Polk, Lancaster City Directory (Lancaster: R.L. Polk & Co., 1905); Lancaster County, Commissioners' Court Minutes, 1916-1920, 5 April 1909, Lancaster County Archives, Lancaster; "Frank H. Shaw is Dead at 78," obituary in Lancaster New Era, 15 May 1950; "Frank H. Shaw Dies; Was Former Water Works Supt," obituary in Lancaster Intelligencer Journal, 15 May 1950; M. Luther Heisey, "The Water Supply of Lancaster" Journal of the Lancaster County Historical Society 41, no. 1 (1937): p. 34.

⁴¹ Justin M. Spivey, "Big Conestoga Creek Bridge No. 12," HAER No. PA-500, Historic American Engineering Record (HAER), National Park Service, U.S. Department of the Interior, August 1998, Prints and Photographs Division, Library of Congress, Washington, D.C., pp. 2, 6.

⁴² The Weaverland Bridge and the Big Chickies Bridge.

⁴³ Spivey.

⁴⁴ Shaw

⁴⁵ "Frank H. Shaw is Dead at 78."

Percy A. Shaw's article, "A Reinforced Concrete Through Girder Bridge," over Conestoga Creek is a valuable piece of descriptive evidence explaining the reasoning underlying the selection of the bridge type and description of building techniques employed in construction of the bridge. It can be assumed that comparable approaches were used in the later bridges designed and supervised by the Shaw firm.

Percy's designed Conestoga girder bridge can shed some light on the Big Chickies Bridge bow arch built six years later at Lancaster Junction. The Conestoga girder and the Big Chickies Bridge slab deck show similarities. The descriptions of how the bridge was reasoned and built are worth discussing here to give us an insight into the early years of reinforced concrete bridge building in Lancaster County as envisioned and practiced by the Shaws. Some of their evolving designs and building methods were probably practiced by the firm throughout the years they designed bridges and supervised their construction.

a. Type and Style of the Conestoga Bridge

For the Conestoga Bridge Percy chose a single clear span (63' x 18') roadway instead of two girder spans "in order to secure the largest possible waterway, and thus provide against the stoppage of drift brought down by the stream during floods, which are frequent in that vicinity."⁴⁶ Abutment and wingwalls from the old bridge were utilized by reinforcing them with a shell of reinforced concrete 12" to 16" thick. Parapet walls were cast monolithic with the wingwalls, but not reinforced. They have deep panels to save weight and to reduce the cost of form work as Percy reasoned that the "span depends for whatever beauty it possesses on the bold, clean lines of the massive concrete work."⁴⁷

b. Bridge Movement

Movement dynamics of the bridge consisting of expansion and contraction due to temperature changes and setting of the concrete were allowed for by resting the span at each end on a half-inch of Hydrex felt and compound while the girders and parapet walls had a heavy coating of the compound between them.

c. Stress Formulas and Concrete Mixtures

⁴⁶ Shaw, p. 744.

⁴⁷ Ibid.

Stress limits and concrete mixtures for the bridge were employed as follows:

Table 2. Conestoga Bridge Formulas		
Stress limits:	Live load:	100 lb. per sq. ft. (equivalent to: 15-ton road roller with axles 11" apart, 6 tons on a 4' wide front wheel, and 4.5 tons on each rear 20" wheel 5' apart center to center).
	Dead load:	weight of bridge plus 9" of crushed stone on the roadway.
	Unit stress:	16,000 lb. in tension 12,000 lb. in compression.
	Concrete compressive strength:	600 lb.
Concrete mixtures:	spans and wingwalls:	1:2:4
	abutments and wingwall foundations:	1:3:6

d. The Crew

The contractor employed one foreman, one assistant foreman, one engineer, two to four carpenters, and five to twelve laborers.

e. Sequence of Construction and Formwork

First the abutment and wingwall reinforcing shell was placed, then the parapet walls, and finally the formwork for the span was erected before the concrete was poured. Timbers, securely braced to prevent washing out and resting on the ledge in the streambed, were used for supports for the span framework.

f. Concrete Work

This crew mixed the concrete by a batch mixer mounted on a truck with a boiler and engine. The reinforcement bars in the span, with the exception of the compression bars, were put in place and securely wired. The concrete was then poured in one continuous operation for the entire span—both floors and girders—in layers from one end of the bridge to the other with no concrete having set when new work was set upon it. The pouring of the concrete ran for fifty-six continuous hours starting at 7 a.m. on September 2 and ended on September 4 in the afternoon. After forty-eight hours, all the forms, with the exception of those supporting the heavy weight of the span, were removed and the concrete surfaces given a final finish. After three weeks, the forms supporting the span were finally removed. Construction of the Conestoga Bridge

commenced June 27, 1910, and was completed September 13, 1910, a total of sixty-six days of construction work.

g. Cost of the Conestoga Bridge to Lancaster County

Table 3 below lists the work items contracted by the County of Lancaster with the Shaw construction company for the building of the Conestoga reinforced concrete bridge. It is assumed that labor costs are subsumed in the prices charged by the Shaw firm for materials and in removing the old bridge and excavation work. The most expensive materials contained in a reinforced concrete bridge are concrete and reinforcement steel. The cost of the concrete and steel bars came to \$3,985.88, or 92 percent of the total cost budgeted by the county for this bridge.

It can be seen then that Percy's information about bridge building—the reasoning involved in choosing the bridge type, the descriptions of construction methods, and a break down of the costs involved—gives us a window into the history of reinforced concrete construction in the early 1900s in Pennsylvania.

Table 3. Conestoga Bridge Construction Items and Costs			
Construction Items	Units	Cost per Unit	Total
Removing old bridge			\$125.00
Concrete (1:2:4)	241.4 cu.yd.	\$11.50	\$2,776.10
Concrete (1:3:6)	49.3 cu.yd.	\$9.00	\$443.70
Steel reinforcing	25,536 lb.	\$0.03	\$766.08
Above mean low water earth excavation	69.5 cu.yd.	\$0.45	\$31.28
Below mean low water earth excavation	10.5 cu.yd.	\$2.10	\$22.05
Rock excavation	38 cu.yd.	\$1.60	\$60.80
Crushed stone for road bed	40 cu.yd.	\$1.60	\$64.00
Wainwright curbing	140 ft.	\$0.206	\$28.82
Name stone			\$10.00
Extra work			\$6.74
GRAND TOTAL			\$4,334.57

Patents and Local Access to Bridge Building

To explain the relationship between Frank H. Shaw and state agencies that contracted him and his bridge designs, we have to understand the historic context—the history of patents

and the process of selecting a particular bridge to be built—that eventually led to the system of favored relationships that Shaw enjoyed. It is necessary to understand the debate going on in the United States at the time when concrete bridges were feverishly being built. The struggle in the early years actually was between standardized design presented by independent contractors, such as Daniel B. Luten, and state engineering offices where engineers produced their own designs.

Starting in the 1890s in the United States, patents for concrete bridges began to be secured especially by major holders of concrete bridge patents. For example in the 1890s, patents were held by Joseph Melan, Fritz von Emperger, Edwin Thacher, Indiana's Daniel B. Luten, who had over forty by 1916, and James B. Marsh, who patented his unique concrete bowstring bridge. These patents were for different styles and what ensued was a debate about how bridges should be built in the United States, should they be offered by independent companies or by governmental offices that determined standard types, shapes and styles?

A number of methods were used in the early days to pick the best bridges. Before an impartial system of bridge selection evolved, a debate raged in America pitting the independent contractor and his design against the state's designers. In a few words, the state wanted to produce standardized designs usually dominated by a few engineers employed by the state, while independent contractors complained that they and their patented designs were being shut out of the business of bridge design and construction.

Engineers working for the government drafted bridges with detailed and standardized specifications that left almost no room for the designer and in the long run for any innovation. This method produced bridges that were costly, excessively heavy, but were designed to counteract the unaccomplished job skills of companies whose low bids won the contract.

Another method employed in Europe picked a bridge design that showed the best planning knowledge and building techniques but which did not necessarily come from the lowest bidder. Unfortunately, this approach did not always guarantee that those evaluating the best engineered design were technically qualified to do so or that governmental officials were not motivated by corruption to be partial.

What eventually developed in the United States was a method that Americans believed emphasized ingenuity in design and construction resulting in a bridge that supposedly was better

and cheaper. This method had the following requirements that were prepared by the regional engineer: a mandatory general plan setting out how a bridge should perform by presenting sets of specifications for loadings and capacities, external dimensions, and the structure's principal architectural features. These specifications, dimensions and features applied to all bidders. The bidders, in reference to the mandatory plan of the regional authority, were required to submit drawings specifying the thickness of bridge parts, and detailing reinforcement schemes. The contractors were also required to guarantee their work, the construction materials used, and their design for one year. The whole process was meant to produce an economical and expertly designed and built bridge.

Consequently, this method produced a unique relationship of consultation between elected officials, the regional engineer, and the contractor. In the case of Frank H. Shaw, it developed into a special relationship between an authorizing governmental agency and the county engineer as the preferred provider of bridge designs. The relationship suggests that he held a monopoly over whose designs the county would choose. Further research into this subject should be carried out to show what percentage of bridges was built by Shaw as compared to his competitors. Shaw's status developed from the special relationship with the county that gave him a unique opportunity to create a historic legacy of bridge design and construction that exists to this day. Through such a close and favored relationship that the firm of Frank H. Shaw enjoyed with the county, his monopoly may have limited access to the market by other contracting firms wishing to make their contributions to the infrastructure of Lancaster County through new designs and construction methods.

In any case, the Big Chickies Creek was hailed as a rarity in 1931 when Lancaster's The Sunday News presented a photograph of a relatively new looking bridge fifteen years after it was built, stating it was an: "Artistic bridge. This unusual concrete structure spans the Little Chiques Creek [which was an error since it spanned Big Chickies Creek] between Sporting Hill and Lancaster Junction and is one of the few of its type in Lancaster County."⁴⁸

SIGNIFICANCE

⁴⁸ The Sunday News, Lancaster, PA, August 30, 1931.

The tied-through reinforced concrete arch Big Chickies Bridge, built in 1916, is a historically and technologically significant early and rare example of just two found in the Commonwealth of Pennsylvania of a rainbow type tied design bridge with diagonal members. Perhaps in attempting to avoid any problems with patents issued to J.B. Marsh, the Shaws created a concrete bridge that has unique features found in the previous era's iron bridges with their diagonal stress elements in the arch. Although the Big Chickies Bridge has not been previously evaluated for National Register Eligibility, the Pennsylvania Historic and Museum Commission did determine in 1993 that its exact twin in equally good condition at Weaverland was eligible for the National Register.

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